

HIGHER DERIVATIVES OF THE GRAM FUNCTION

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ABSTRACT. We find the asymptotics of higher-order derivatives of the Gram function t_u . We also prove the uniform distribution and continuous uniform distribution modulo one of t_u^n for $n = 1, 2, \dots$ and make a conjecture that the Riemann zeta function shifted by powers of the Gram points has discrete universality.

1. INTRODUCTION AND MAIN RESULTS

The concept of Gram points was introduced by the Danish mathematician Gram. They are points on the critical line where the Riemann-Siegel theta function $\vartheta(t)$ is a multiple of π . Using a heuristic argument, Gram [3] reasoned that there is precisely one zero of the Riemann zeta function between two neighboring Gram points. Later on, Hutchinson [4] found counterexamples to this conjecture. Nevertheless, the Gram points remain an active area of research.

The Riemann-Siegel theta function $\vartheta(t)$ gives us an increment of the continuous branch of the argument of $\pi^{-s/2}\Gamma(s/2)$ as s goes along the straight line segment of the complex plane joining $s = 1/2$ and $s = 1/2 + it$; note that $\vartheta(t)$ is increasing for $t \geq 7$ (Ivić [5, Section 1.3] and Korolev [10, p. 3]). The function ϑ appears in the symmetric functional equation involving the Riemann zeta function ζ

$$e^{i\vartheta(t)}\zeta(1/2 + it) = e^{-i\vartheta(t)}\zeta(1/2 - it).$$

The function ϑ is related to the function S . The function $S(t)$ is defined as

$$S(t) = \frac{1}{\pi} \arg \zeta(1/2 + it).$$

Here, the argument is taken along the continuous branch of $\arg \zeta(s)$ as we move from $s = 2$ to $s = 2 + it$ and $s = 1/2 + it$. If s is a zero of ζ , then we set

$$S(t) = \lim_{\delta \rightarrow +0} \frac{1}{2} (S(t + \delta) + S(t - \delta)).$$

The Riemann-von Mangoldt formula (see, e.g. Edwards [1, Section 6.6], Ivić [5, formula (1.45)], Karatsuba and Korolev [6]) leads to the following relation between the functions ϑ and S

$$(1) \quad N(t) = \frac{1}{\pi} \vartheta(t) + 1 + S(t).$$

Here $N(t)$ is the number of ζ zeros in the rectangle $0 < \Im s \leq t$, $0 \leq \Re s \leq 1$, with the exception that at points of discontinuity, which coincide with the ordinates of

the zeros γ_n , it is defined as the limit

$$N(t) = \frac{1}{2} \lim_{h \rightarrow 0} (N(t+h) + N(t-h)).$$

The function S is related to the function N by the Riemann-von Mangoldt formula.

$$N(t) = \frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8} + S(t) + \frac{1}{\pi} \delta(t)$$

(see Karatsuba and Korolev [6, Theorem 1]).

From the above equation, the function S can be expressed as

$$(2) \quad S(t) = -\frac{t}{2\pi} \log \frac{t}{2\pi} + \frac{t}{2\pi} - \frac{7}{8} - \frac{1}{\pi} \delta(t) + N(\gamma + 0)$$

with $\gamma < t < \gamma'$, where γ and γ' are neighboring ordinates of the zeros of ζ (see Karatsuba and Korolev [6, Proof of Theorem 2]). The function δ is

$$(3) \quad \delta(t) = \frac{t}{4} \log \left(1 + \frac{1}{4t^2} \right) + \frac{1}{4} \arctan \frac{1}{2t} - \frac{t}{2} \int_0^\infty \frac{\rho(u) du}{(u + 1/4)^2 + (t/2)^2}$$

and $\rho(u) = 1/2 - \{u\}$ (see Karatsuba and Korolev [6, Theorem 1]).

The first derivative of S is

$$(4) \quad S'(t) = -\frac{1}{2\pi} \log \frac{t}{2\pi} + O(t^{-2}), \quad t \rightarrow +\infty$$

except for its points of discontinuity (see Karatsuba and Korolev [6, Theorem 2]).

We give our theorem about the n -th derivative of the S function.

Theorem 1. *The n -th derivative of the S function when $n \geq 2$ is*

$$(5) \quad S^{(n)}(t) = (-1)^{n-1} \frac{(n-2)!}{2\pi} t^{1-n} + O(t^{-n-1}), \quad t \rightarrow +\infty.$$

except for its discontinuity points.

From the above theorem, equation (1), equation (4), and the fact that ϑ is differentiable infinitely many times, we get

Corollary 2. *Suppose $n \geq 2$. Then, the n -th derivative of the Riemann-Siegel theta function is*

$$(6) \quad \vartheta^{(n)}(t) = (-1)^n \frac{(n-2)!}{2} t^{1-n} + O(t^{-n-1}), \quad t \rightarrow +\infty.$$

It is known that for $t \geq 7$, the Riemann-Siegel theta function $\vartheta(t)$ is monotonically increasing. Thus for $u \geq \vartheta(7)/\pi + \pi$, we can implicitly define the Gram function t_u by

$$(7) \quad \vartheta(t_u) = (u-1)\pi,$$

with $t_u \geq 7$. Note that for positive integers k , t_k correspond to Gram points (see Edwards [1, pp. 125–226]).

The asymptotic expression for the Gram function is (see Lavrik [14, Lemma 2])

$$(8) \quad t_u = \frac{2\pi u}{\log u} \left(1 + (1 + o(1)) \frac{\log \log u}{\log u} \right), \quad u \rightarrow +\infty.$$

Korolev [10, Lemma 1.1] gave the asymptotics for the first two derivatives of the Gram function. The first derivative of the Gram function is

$$(9) \quad \frac{dt_u}{du} = \frac{2\pi}{\log u} \left(1 + (1 + o(1)) \frac{\log \log u}{\log u} \right),$$

as $u \rightarrow +\infty$.

We calculate the n -th derivative of the Gram function t_u , which we denote by $t_u^{(n)}$.

Theorem 3. *The n -th derivative of the Gram function is asymptotically*

$$t_u^{(n)} = \frac{(-1)^{n+1} 2\pi(n-2)!}{u^{n-1} \log^2 u} \left(1 + (2 + o(1)) \frac{\log \log u}{\log u} \right), \quad u \rightarrow +\infty,$$

for $n = 2, \dots$

Here, we corrected the constant in [10, Lemma 1.1] for $n = 2$.

The proof of the following theorem of the uniform distribution modulo one of the sequence $\{t_k^n\}_{k=1}^\infty$ is analogous to Pańkowski [17, Proof of Theorem 1].

Theorem 4. *The sequence $\{t_k^n\}_{k=1}^\infty$ is uniformly distributed modulo one for $n = 1, 2, 3, \dots$*

We have the following corollary about the continuous uniform distribution of the n -th degree of the Gram function. This result was proved by Laurinćikas [13, Lemma 5.7] by a different method. In our case, it follows directly from Theorem 4.

Corollary 5. *The function t_u^n , $n = 1, 2, 3, \dots$, is continuously uniformly distributed modulo one.*

The function t_u can be applied in the universality theory of zeta functions. As an example, we will consider the Riemann zeta function. Recall that the universality of $\zeta(s)$ in the Voronin sense means that a broad class of analytic functions can be approximated by shifts $\zeta(s + iu)$, $u \in \mathbb{R}$. This approximation property of $\zeta(s)$ was discovered by Voronin [20] and has various theoretical and practical applications. For results, see Matsumoto [15].

Now we discuss the discrete universality for the Riemann zeta function using its shifts by the degrees of the Gram points $\zeta(s + it_k^n)$. This problem has been extensively studied by Korolev and Laurinćikas. In [7], they proved it for the degree 1. They proved the result for short intervals in [8]. The continuous version of joint universality was proved in [9].

By $\#A$ we denote the cardinality of the set A . We make the assumption about the second moment of the Riemann zeta function shifted by the Gram function: For all $\epsilon > 0$ and sufficiently large y and N , we have

$$(10) \quad \frac{1}{N} \sum_{k=2} |\zeta(s + it_k^n) - \zeta_{\{p:p \leq y\}}(s + it_k^n)|^2 < \epsilon$$

with $s \in K$ where K is any compact set with connected complement in the right part of the critical strip. In addition, $\zeta_{\{p:p \leq y\}}(s) := \prod_{p \leq y} (1 - 1/p^s)^{-1}$ is a truncated Euler product for the Riemann zeta function, where p is prime. Pańkowski [17, p. 192, Proposition 1] incorrectly thought that this follows from Gallagher's lemma applied to Dirichlet L -functions with shifts $\alpha t^a (\log t)^b$ for $\alpha \neq 0$, $a > 0$, and b real numbers. Sourmelidis [19] partially remedied this error. The proof of universality follows from our assumption (10) and Theorem 4 using the method by Pańkowski [17, pp. 192–194, proof of Theorem 2]. We formulate our statement as follows: *Suppose, $K \subset \{s \in \mathbb{C} : 1/2 < \sigma < 1\}$ is a compact set with connected complement, f is a continuous non-vanishing function on K , analytic inside of K . Then, for*

any $\epsilon > 0$,

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \# \left\{ 2 \leq k \leq N : \max_{s \in K} |\zeta(s + it_k^n) - f(s)| < \epsilon \right\} > 0.$$

Let $D = \{s \in \mathbb{C} : 1/2 < \sigma < 1\}$. Denote by $\mathcal{K}$ the class of compact subsets of the strip D with connected complements, and by $H_0(K)$ with $K \in \mathcal{K}$ the class of continuous non-vanishing functions in K that are analytic in the interior of K . Let $\text{meas } A$ stand for the Lebesgue measure of a measurable set $A \subset \mathbb{R}$. Denote by $\#A$ the cardinality of the set A , and let $N \in \mathbb{N}$.

Theorem 6. *Suppose that $\alpha \in (0, 1]$. Let $K \in \mathcal{K}$ and $f(s) \in \overline{H_0(K)}$. Then, for every $\epsilon > 0$,*

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \# \{1 \leq k \leq N : \sup_{s \in K} |\zeta(s + it_k^\alpha) - f(s)| < \epsilon\} > 0.$$

Moreover, for all but at most countably many ϵ , “ $\liminf$ ” can be replaced by “ $\lim$ ”.

2. PROOFS

Proof of Theorem 1. Our goal is to compute the n -th derivative of $S(t)$. Recall the formula (2). We split the computation into two parts.

First, let us compute

$$\left(-\frac{t}{2\pi} \log \frac{t}{2\pi} + \frac{t}{2\pi} - \frac{7}{8} \right)^{(n)} = \left(-\frac{1}{2\pi t} \right)^{(n-2)} = (-1)^{n-1} \frac{(n-2)!}{2\pi t^{n-1}}.$$

It remains to estimate the error term arising from differentiating the function δ . The terms

$$(11) \quad \frac{t}{4} \log \left(1 + \frac{1}{4t^2} \right) + \frac{1}{4} \arctan \frac{1}{2t}$$

can be expanded into

$$\frac{1}{16t} + \frac{1}{8t} + \sum_{n=3}^{\infty} \frac{a_n}{t^n}$$

with real coefficients a_n . Differentiating the above expression n times term-by-term yields

$$(12) \quad (-1)^n \frac{3n!}{16t^{n+1}} + O(t^{-n-3}), \quad t \rightarrow +\infty.$$

For our purposes, the accuracy $O(t^{-n-1})$ is sufficient.

Continuing our computation of the n -th derivative of $\delta(t)$, let us denote

$$j(t) := \int_0^\infty \frac{\rho(u) du}{(u + 1/4)^2 + (t/2)^2},$$

$$\sigma(u) := \int_0^u \rho(z) dz.$$

We compute the n -th derivative of j . We follow with integration by parts

$$j(t) = \int_0^\infty \frac{d\sigma(u)}{(u + 1/4)^2 + (t/2)^2} = 2 \int_0^\infty \frac{\sigma(u)(u + 1/4) du}{((u + 1/4)^2 + (t/2)^2)^2}.$$

Note that $0 \leq \sigma(u) \leq 1/8$.

We have to differentiate the function $j(t)$ with respect to t n times, which amounts to differentiation under the integral sign. We use the Leibniz rule

$$\frac{d}{dt} \int_a^\infty f(u, t) du = \int_a^\infty \frac{\partial}{\partial t} f(u, t) du.$$

The following conditions must be satisfied: $f(u, t)$ and $\partial f / \partial t$ be continuous for $u \geq a$, and there exists an integrable $g(u)$ such that $|\partial f(u, t) / \partial t| \leq g(u)$ (see Rudin [18, pp. 180–182]). Evidently, they hold in our case.

Applying the Leibniz rule yields

$$\begin{aligned} \frac{d^n}{dt^n} \int_0^\infty \frac{(u + 1/4)\sigma(u) du}{((u + 1/4)^2 + (t/2)^2)^2} &= \int_0^\infty \frac{\partial^n}{\partial t^n} \frac{(u + 1/4)\sigma(u) du}{((u + 1/4)^2 + (t/2)^2)^2} \\ &\ll \int_0^\infty \frac{t^n(4u + 1)\sigma(u) du}{((4u + 1)^2 + 4t^2)^{n+2}}, \quad t \rightarrow +\infty. \end{aligned}$$

Let us split the above integral into two

$$\int_0^t \frac{t^n(4u + 1)\sigma(u) du}{((4u + 1)^2 + 4t^2)^{n+2}} + \int_t^\infty \frac{t^n(4u + 1)\sigma(u) du}{((4u + 1)^2 + 4t^2)^{n+2}}.$$

Let us integrate with respect to u using the facts that $u < t$ in the first integral and $u > t$ in the second. Combining this with the bounds $0 \leq \sigma(u) \leq 1/8$, gives the estimate

$$\begin{aligned} j^{(n)}(t) &\ll \int_0^t \frac{t^n(4u + 1)\sigma(u) du}{((4u + 1)^2 + 4u^2)^{n+2}} + \int_t^\infty \frac{u^n(4u + 1)\sigma(u) du}{(4u + 1)^{2n+4}} \\ &\ll t^{-n-2}, \quad t \rightarrow +\infty. \end{aligned}$$

We need to find the n -th derivative of the last term of the function $\delta(t)$ (see (3))

$$-\frac{t}{2} \int_0^\infty \frac{\rho(u) du}{(u + 1/4)^2 + (t/2)^2} = -\frac{t}{2} j(t).$$

We apply the Leibniz rule to differentiate products. If u and v are n times differentiable functions of x , then (see Gradshteyn and Ryzhik [2, Formula 0.42])

$$\frac{d^n(uv)}{dx^n} = u \frac{d^n v}{dx^n} + \binom{n}{1} \frac{du}{dx} \frac{d^{n-1} v}{dx^{n-1}} + \binom{n}{2} \frac{d^2 u}{dx^2} \frac{d^{n-2} v}{dx^{n-2}} + \dots + \frac{d^n u}{dx^n} v.$$

We get

$$-\frac{1}{2} \left(t j^{(n)}(t) + \binom{n}{1} j^{(n-1)}(t) \right) = O(t^{-n-1}).$$

Collecting this and (12) completes the proof of Theorem 1. $\square$

Proof of Theorem 3. We prove this by induction. Let us make the inductive hypothesis

$$(13) \quad t_u^{(k)} = \frac{(-1)^{k+1} 2\pi(k-2)!}{u^{k-1} \log^2 u} \left(1 + (2 + o(1)) \frac{\log \log u}{\log u} \right), \quad u \rightarrow +\infty.$$

First, we prove that the theorem statement holds for $k = 2$. Differentiating the equation (7) gives

$$\vartheta''(t_u) t_u'^2 + \vartheta'(t_u) t_u'' = 0.$$

We calculate the term $\vartheta''(t_u)t_u'^2$ using Corollary 2, equations (9) and (8)

$$\begin{aligned}\vartheta''(t_u)t_u'^2 &= \left(\frac{1}{2t_u} + O(t_u^{-3})\right) \left(\frac{2\pi}{\log u} \left(1 + (1 + o(1)) \frac{\log \log u}{\log u}\right)\right)^2 \\ &= \frac{2\pi}{2u \log u} \left(1 + (1 + o(1)) \frac{\log \log u}{\log u}\right), \quad u \rightarrow +\infty.\end{aligned}$$

Differentiating (7) and using the expression (9) yields

$$\frac{1}{\vartheta'(t_u)} = \frac{2}{\log u} \left(1 + (1 + o(1)) \frac{\log \log u}{\log u}\right), \quad u \rightarrow +\infty.$$

Combining the above two results gives

$$t_u'' = -\frac{2\pi}{u \log^2 u} \left(1 + (2 + o(1)) \frac{\log \log u}{\log u}\right), \quad u \rightarrow +\infty.$$

Suppose our hypothesis (13) holds for $2 \leq k \leq n-1$. We will show that it does for $k = n$. By Faà di Bruno's formula (see Gradshteyn and Ryzhik [2, formula (0.430)]), the n -th derivative of a composite function $h(x) = g(f(x))$ is

$$(14) \quad \frac{d^n}{dx^n} g(f(x)) = \sum \frac{n!}{m_1! 1!^{m_1} m_2! 2!^{m_2} \dots m_n! n!^{m_n}} g^{(m_1+m_2+\dots+m_n)}(f(x)) \\ \times \prod_{j=1}^n (f^{(j)}(x))^{m_j},$$

here, the summation is taken over all partitions

$$(15) \quad m_1 + 2m_2 + \dots + nm_n = n.$$

We will need the shorthand notation $\sum m = m_1 + m_2 + \dots + m_n$ for our purposes.

We plug $g(s) = \vartheta(s)$, $f(u) = t_u$ in Faà di Bruno's formula. Then, the relation (7) yields that the $d^n \vartheta(t_u)/du^n$ equals 0 and the formula (14) becomes

$$(16) \quad 0 = \vartheta'(t_u)t_u^{(n)} + \dots + \vartheta^{(n)}(t_u)(t_u')^n.$$

The first member in the above equation corresponds to the case $m_n = 1$ and $m_i = 0$ for $i = 1, \dots, n-1$. The last member - to the case $m_1 = n$ and $m_i = 0$ for $i = 2, \dots, n$.

In the summation of Faà di Bruno's formula, the first term in the above equation (16) is the only one containing $t_u^{(n)}$, and the last one is asymptotically the largest of the remaining ones as $u \rightarrow \infty$. Let us prove this fact. We will show that $\vartheta^{(n)}(t_u)(t_u')^n$ is asymptotically larger than the terms obtained using Faà di Bruno's formula, except possibly the first one. Hence, we can ignore the constants. Any term is uniquely determined by the values $m_1, m_2, \dots, m_n$, which come from the partition (15). For this calculation, we use the asymptotics for the derivatives of $\vartheta(t)$ (Corollary 2), the asymptotics for t_u' and t_u , and our inductive step (13). We skip the term $\vartheta'(t_u)t_u^{(n)}$. In all the remaining terms, we have $m_n = 0$. The arbitrary term in Faà de Bruno's formula, except for the partition $m_n = 1, m_i = 0$,

$i = 1, \dots, n-1$, is

$$\begin{aligned}
 \vartheta^{(m_1+m_2+\dots+m_{n-1})}(t_u) \prod_{j=1}^{n-1} (t_u^{(j)})^{m_j} &= \Theta \left(\frac{1}{t_u^{\sum m-1}} \cdot \frac{1}{\log^{m_1} u} \prod_{j=2}^{n-1} \left(\frac{1}{u^{j-1} \log^2 u} \right)^{m_j} \right) \\
 &= \Theta \left(\frac{\log \sum^{m-1} u}{u^{\sum m-1} \log^{m_1} u} \cdot \frac{1}{u^{\sum_{j=2}^{n-1} j m_j - \sum m + m_1} \log^2 \sum^{m-2m_1} u} \right) \\
 (17) \quad &= \Theta \left(\frac{1}{u^{n-1} \log \sum^{m-m_1+1} u} \right).
 \end{aligned}$$

We say that a function $f(x)$ is $\Theta(g(x))$ if there exist positive constants c_1, c_2 and a real number x_0 such that $c_1 g(x) \leq f(x) \leq c_2 g(x)$, $x \geq x_0$. In the last step above, we used the equation (15). The last expression shows that among all the terms in Faà de Bruno's formula, except $\vartheta'(t_u) t_u^{(n)}$, the asymptotically largest one is $\vartheta^{(n)}(t_u) (t_u')^n$, corresponding to the partition $m_1 = n$ and $m_i = 0$, $i = 2, \dots, n$. This, together with (16), leads to

$$\vartheta'(t_u) t_u^{(n)} = -\vartheta^{(n)}(t_u) (t_u')^n + o\left(\vartheta^{(n)}(t_u) (t_u')^n\right) \quad (u \rightarrow \infty).$$

$$\vartheta'(t_u) t_u^{(n)} = -\vartheta^{(n)}(t_u) (t_u')^n + O\left(\frac{1}{u^{n-1} \log^2 u}\right) \quad (u \rightarrow \infty),$$

It follows

$$t_u^{(n)} = \frac{-\vartheta^{(n)}(t_u) (t_u')^n}{\vartheta'(t_u)} + O\left(\frac{1}{\vartheta'(t_u) u^{n-1} \log^2 u}\right).$$

Corollary 2 together with the asymptotics for t_u' and t_u leads to

$$t_u^{(n)} = \frac{(-1)^{n+1} 2\pi(n-2)!}{u^{n-1} \log^2 u} \left(1 + (2 + o(1)) \frac{\log \log u}{\log u} \right).$$

This completes the induction and yields the theorem. $\square$

Proof of Theorem 4. We have theorem [11, Chapter 1, Theorem 3.5] stating: if there exists $l \in \mathbb{N}$, the function $f(x)$ is defined for $x \geq 1$, the derivative $f^{(l)}(x)$ tends monotonically to 0 as $x \rightarrow \infty$, and if $x|f^{(l)}(x)| \rightarrow \infty$, $x \rightarrow \infty$, then the sequence $\{f(n)\}_{n \in \mathbb{N}}$ is uniformly distributed modulo one.

Consider the n -th degree of the Gram function t_u^n . By Faà di Bruno's formula, we have

$$(18) \quad \frac{d^n t_u^n}{du^n} = \sum \frac{n!}{m_1! 1!^{m_1} m_2! 2!^{m_2} \dots m_n! n!^{m_n}} \frac{d^{m_1+m_2+\dots+m_n} t_u^n}{dt_u^{m_1+m_2+\dots+m_n}} \prod_{j=1}^n (t_u^{(j)})^{m_j},$$

where we sum over all partitions

$$(19) \quad m_1 + 2m_2 + \dots + nm_n = n.$$

Let us denote $\sum m := m_1 + m_2 + \dots + m_n$. An arbitrary term on the right-hand side of the equation (18) depends on the value of the vector $(m_1, m_2, \dots, m_n)$. Ignoring the constant, recalling the asymptotic of t_u given in the formula (8), and

the asymptotic of its derivative in Theorem 3 yields the following asymptotic of an arbitrary term in the formula (18)

$$\begin{aligned} & \frac{u^{n-\sum m}}{\log^{n-\sum m} u} \cdot \frac{1}{\log^{m_1} u} \prod_{j=2}^n \frac{1}{(u^{j-1} \log^2 u)^{m_j}} = \frac{u^{n-\sum m}}{\log^{n-\sum m} u} \cdot \frac{1}{u^{n-\sum m} \log^{-m_1+2\sum m} u} \\ & = \frac{1}{\log^{n-m_1+\sum m} u}, \end{aligned}$$

where we used the formula (19). This value is maximized when we minimize $-m_1 + \sum m$, which happens when $m_1 = n$ and $m_2 = \dots = m_n = 0$.

To prove monotonicity, we calculate the $n+1$ -st derivative of the Gram function. Again, we use Faà di Bruno's formula. Summing over partitions, we have

$$m'_1 + 2m'_2 + \dots + nm'_n + (n+1)m'_{n+1} = n+1,$$

where the integers $m'_1, m'_2, \dots, m'_{n+1}$ appear in our formula now. Let us denote $\sum m' := m'_1 + 2m'_2 + \dots + m'_{n+1}$. It follows

$$\begin{aligned} & \frac{u^{n-\sum m'}}{\log^{n-\sum m'} u} \cdot \frac{1}{\log^{m'_1} u} \prod_{j=2}^{n+1} \frac{1}{(u^{j-1} \log^2 u)^{m'_j}} = \frac{u^{n-\sum m'}}{\log^{n-\sum m'} u} \cdot \frac{1}{u^{n+1-\sum m'} \log^{-m'_1+2\sum m'} u} \\ & = \frac{1}{u \log^{n-m'_1+\sum m'} u} \end{aligned}$$

Again, this value is maximized by setting $m'_1 = n+1$, $m'_2 = m'_3 = \dots = m'_{n+1} = 0$.

All in all, we have

$$\frac{d^n t_u^n}{du^n} = \Theta\left(\frac{1}{\log^n u}\right)$$

and

$$\frac{d^{n+1} t_u^n}{du^{n+1}} = \Theta\left(\frac{1}{u \log^n u}\right).$$

Based on the continuity of the $n+1$ -st derivative, there exists some value u_0 for which $d^{n+1} t_u^n / du^{n+1}$ does not change its sign provided $u > u_0$. Hence, the n -th derivative of t_u^n will monotonically approach 0 for $u > u_0$. In addition, $u |d^n t_u^n / du^n|$ will monotonically approach $+\infty$. Therefore, by the criterion stated at the beginning of the proof, the sequence $\{t_k^n\}_{k=1}^\infty$ is uniformly distributed modulo one. $\square$

Proof of Corollary 5. By Kuipers and Niederreiter [11, Chapter 1, Theorem 9.6], the function $f(t)$, $t \geq 0$, is continuously uniformly distributed modulo one if 1) the sequence $f(k+t)$, $k = 0, 1, \dots$, is uniformly distributed modulo one for almost all $t \in [0, 1)$ and 2) the sequence $f(kt)$, $k = 1, 2, \dots$, is uniformly distributed modulo one for almost all $t \in [0, 1)$. If we take f to be the n -th degree of the Gram function, both of these facts follow from the fact that there exists real number u_0 such that $d^n t_u^n / du^n$ monotonically approaches zero for $u > u_0$, $u |d^n t_u^n / du^n|$ monotonically approaches $+\infty$ for $u > u_0$, and the criterion for the uniform distribution modulo 1 of sequences, formulated at the beginning of the proof of Theorem 4. $\square$

3. PROBABILISTIC MODEL

Denote by $H(D)$ the space of analytic functions on D endowed with the topology of uniform convergence on compacta. Proof of Theorem ?? is based on a limit

theorem of weakly convergent probability measures in the space $H(D)$. For its statement, we need some notation and definitions. Define the set

$$\Omega = \prod_p \gamma_p,$$

where the Cartesian product is taken over all primes, and $\gamma_p = \{s \in \mathbb{C} : |s| = 1\}$. With the product topology and point-wise multiplication, the torus Ω is a compact topological Abelian group, therefore on $(\Omega, \mathcal{B}(\Omega))$ ($\mathcal{B}(\mathbb{X})$ is the Borel σ -field on the space $\mathbb{X}$), the probability Haar measure m_H can be defined. This leads to the probability space $(\Omega, \mathcal{B}(\Omega), m_H)$. Denote by $\omega(p)$ the p -th component of an element $\omega \in \Omega$, and on $(\Omega, \mathcal{B}(\Omega), m_H)$ define the $H(\Omega)$ -valued random element

$$\xi(s, \omega) = \prod_p \left(1 - \frac{\omega(p)}{p^s}\right)^{-1}.$$

Note that the latter infinite product, for almost all $\omega \in \Omega$, is uniformly convergent on compact subsets of the strip D . Let D_ξ denote the distribution of $\xi(s, \omega)$.

For $A \in \mathcal{B}(H(D))$

$$P_T(A) = \frac{1}{T} \text{meas}\{u \in [0, T] : \zeta(s + it_k^\alpha) \in A\}.$$

The main ingredient of proofs of Theorems ?? and 6 is the following two limit theorems.

Proposition 7. *Suppose that α is an arbitrary positive number. Then P_T converges weakly to P_ξ as $T \rightarrow \infty$.*

Similarly, define

$$Q_N(A) = \frac{1}{N} \#\{1 \leq k \leq N : \zeta(s + it_k^\alpha) \in A\}, \quad A \in \mathcal{B}(H(D)).$$

Then we have

Proposition 8. *Suppose that $\alpha \in (0, 1]$. Then Q_N converges weakly to P_ξ as $N \rightarrow \infty$.*

We start the proofs of Theorems 7 and 8 with limit theorems on the torus Ω . Denote by $\mathbb{P}$ the set of all prime numbers. For $A \in \mathcal{B}(\Omega)$, define

$$P_T^\Omega(A) = \frac{1}{T} \text{meas}\{u \in [0, T] : (p^{-it_u^\alpha} : p \in \mathbb{P}) \in A\}.$$

Lemma 9. *Suppose that α is an arbitrary positive fixed number. Then P_T^Ω converges weakly to the Haar measure m_H as $T \rightarrow \infty$.*

Proof. Denote by $g_T(\underline{l})$, $\underline{l} = (l_p : l_p \in \mathbb{Z}, p \in \mathbb{P})$, be the Fourier transform of P_T^Ω , i.e.

$$\begin{aligned} g_T(\underline{l}) &= \int_\Omega \left(\prod_{p \in \mathbb{P}}' \omega^{l_p}(p) \right) dP_T^\Omega = \frac{1}{T} \int_0^T \left(\prod_{p \in \mathbb{P}}' p^{-il_p t_u^\alpha} \right) du \\ (20) \quad &= \frac{1}{T} \int_0^T \exp \left\{ -it_u^\alpha \sum_{p \in \mathbb{P}}' l_p \log p \right\} du, \end{aligned}$$

where the sign ' shows that only a finite number of integers l_p are distinct from zero. Obviously,

$$(21) \quad g_T(\underline{0}) = 1.$$

Since the set $\{\log p : p \in \mathbb{P}\}$ is linearly independent over $\mathbb{Q}$,

$$A(\underline{l}) =: \sum'_{p \in \mathbb{P}} l_p \log p \neq 0.$$

for $\underline{l} \neq \underline{0}$. It is well known that if $g(x)$ is a real-valued differentiable function with monotonic derivative and $g'(x) \geq \lambda > 0$ or $g'(x) \leq -\lambda < 0$ in the interval (a, b) , then

$$\int_a^b \exp\{ig(x)\} dx \ll \frac{1}{\lambda}.$$

Therefore, we obtain by (20) and properties of t_u that, for $\underline{l} \neq \underline{0}$, [\[reikia daugiau detaliu surasyti, taip pat kuo remiamasi\]](#)

$$g_T(\underline{l}) = \frac{1}{T} \int_{\sqrt{T}}^T \exp\{-it_u^\alpha A(\underline{l})\} du + o(1) = o(1), \quad T \rightarrow \infty.$$

Thus, in view of (21)

$$\lim_{T \rightarrow \infty} g_T(\underline{l}) = \begin{cases} 1 & \text{if } \underline{l} = \underline{0}, \\ 0 & \text{if } \underline{l} \neq \underline{0}, \end{cases}$$

and this proves the lemma. $\square$

For $A \in \mathcal{B}(\Omega)$, define

$$Q_N^\Omega(A) = \frac{1}{N} \#\{1 \leq k \leq N : (p^{-it_u^\alpha} : p \in \mathbb{P}) \in A\}.$$

Lemma 10. *Suppose that $\alpha \in (0, 1]$. Then Q_N^Ω converges weakly to the Haar measure m_H at $N \rightarrow \infty$.*

Proof. We know that the Fourier transform $q_N(\underline{l})$ of Q_N^Ω is

$$(22) \quad q_N(\underline{l}) = \frac{1}{N} \sum_{k=1}^N \exp \left\{ -it_k^\alpha \sum'_{p \in \mathbb{P}} l_p \log p \right\}.$$

Thus,

$$(23) \quad q_N(\underline{0}) = 1.$$

The function $(t_u^\alpha)'$ goes monotonically to zero as $u \rightarrow \infty$, and $\lim_{u \rightarrow \infty} u(t_u^\alpha)' = \infty$. Therefore, by Theorem 3.5 in [11], the sequence $\{t_k^\alpha a\}$ with every $a \in \mathbb{R} \setminus \{0\}$ is uniformly distributed modulo 1 [\[Pastaruosius du sakinius irgi reikia pagrįsti\]](#). Hence, (22) and the application of the Weyl criterion show that

$$\lim_{N \rightarrow \infty} q_N(\underline{l}) = 0$$

for $\underline{l} \neq \underline{0}$, and this together with (23) proves the lemma. $\square$

For any $n \in \mathbb{N}$, let $v_n(m) = \exp\{-(m/n)^\vartheta\}$, where $\vartheta > 1/2$ is a fixed number, and

$$\zeta_n(s) = \sum_{m=1}^{\infty} \frac{v_n(m)}{m^s}.$$

Then we have, see, for example, [12], that the latter series is absolutely convergent for $\sigma > 1/2$. Restrict the functions $\omega(p)$, $\omega \in \Omega$, $p \in \mathbb{P}$, to the set $\mathbb{N}$ by setting

$$\omega(m) = \prod_{p^l | m, p^2 \nmid m} \omega^l(p), \quad m \in \mathbb{N},$$

and define the series

$$\zeta_n(s, \omega) = \sum_{m=1}^{\infty} \frac{\omega(m) v_n(m)}{m^s}$$

which is also absolutely convergent for $\sigma > 1/2$. Then the $u_n : \Omega \rightarrow H(D)$ given by

$$u_n(\omega) = \zeta(\cdot, \omega)$$

is continuous. Moreover,

$$u_n(p^{-it_u^\alpha} : p \in \mathbb{P}) = \zeta_n(s + it_u^\alpha).$$

Therefore, putting for $A \in \mathcal{B}(H(D))$

$$P_{T,n}(A) = \frac{1}{T} \text{meas}\{u \in [0, T] : \zeta_n(s + it_k^\alpha) \in A\},$$

we have $P_{T,n} = P_T^\Omega u_n^{-1}$ and $Q_{N,n} = Q_N^\Omega u_n^{-1}$. This and Lemmas 9 and 10 lead to the following limit results.

Lemma 11. *Suppose that α is an arbitrary fixed number. Then $P_{T,n}$ converges weakly to measure $V_n =_{\text{def}} m_H u_n^{-1}$ as $T \rightarrow \infty$.*

Lemma 12. *Suppose that $\alpha \in (0, 1]$ is a fixed number. Then $Q_{N,n}$ converges weakly to the measure V_n as $N \rightarrow \infty$.*

We recall that if P is a probability measure on $(\mathbb{X}, \mathcal{B}(\mathbb{X}))$ and $h : \mathbb{X} \rightarrow \mathbb{X}$ is a $(\mathcal{B}(\mathbb{X}), \mathcal{B}(\mathbb{X}))$ -measurable function, then the measure Ph^{-1} is defined by $Ph^{-1}(A) = P(h^{-1}A)$, $A \in \mathcal{B}(\mathbb{X})$.

The measure V_n clearly appears in the investigation of weak convergence for

$$\frac{1}{T} \text{meas}\{\tau \in [0, T] : \zeta(s + i\tau) \in A\}, \quad A \in \mathcal{B}(\cdot(D))$$

as $T \rightarrow \infty$. Therefore, the following important assertion is well-known, see, for example, [12].

Lemma 13. *V_n converges weakly to P_ξ as $n \rightarrow \infty$. Moreover, the support of P_ξ is the set*

$$S =_{\text{def}} \{g \in H(D) : g(s) \neq 0 \text{ or } g(s) \equiv 0\}$$

From Lemma 13, it follows that, for the proof of Theorems ? and ?, it suffices to show that the limit measures of P_T and Q_N coincide with that of V_n . For this, we need some ? square estimates

Lemma 14. *Suppose that α is an arbitrary fixed positive number, and $t \in \mathbb{R}$. Then, for fixed $\sigma \in (1/2, 1)$,*

$$\int_1^T |\zeta(\sigma + it + it_u^\alpha)|^2 du \ll_{\sigma, \alpha} T(1 + |t|).$$

Proof. It is well known that, for $\sigma \in (1/2, 1)$

$$(24) \quad \int_1^T |\zeta(\sigma + iu)|^2 du \ll_{\sigma} T.$$

Therefore, we have

$$\int_x^{2x} |\zeta(\sigma + it + it_u^\alpha)|^2 du = \int_x^{2x} \frac{1}{(t_u^\alpha)}$$

?

$$\begin{aligned} \max_{X \leq u \leq 2X} \frac{1}{(t_u^\alpha)'} \int_X^{2X} d \left(\int_1^{t+t_u^\alpha} |\zeta(\sigma + iv)|^2 dv \right) &\leq \max_{X \leq k \leq 2X} \frac{1}{(t_u^\alpha)'} \int_1^{|t|+t_u^\alpha} |\zeta(\sigma + iv)|^2 dv \Big|_X^{2X} \\ &\ll_{\sigma} \max_{X \leq k \leq 2X} \frac{1}{(t_u^\alpha)'} (|t| + t_{2X}^\alpha) \ll_{\sigma, \alpha} X(|t| + 1). \end{aligned}$$

Taking $X = T2^{-k}$ and summing over $k = 0, 1, \dots$, gives the assertion of the lemma. $\square$

Lemma 15. *Suppose that $\alpha \in (0, 1]$ is a fixed number and $t \in \mathbb{R}$. Then, for fixed $\sigma \in (1/2, 1)$,*

$$\int_1^T ((t_u^\alpha)')^2 |\zeta'(\sigma + it + it_u^\alpha)|^2 du \ll_{\sigma, \alpha} T(1 + |t|).$$

Proof. It is well known that, for $\sigma \in (1/2, 1)$

$$\int_1^T |\zeta'(\sigma + iu)|^2 du \ll_{\sigma} T.$$

Therefore, we have

$$\begin{aligned} \int_X^{2X} ((t_u^\alpha)')^2 ((t_u^\alpha)')^2 |\zeta'(\sigma + it + it_u^\alpha)|^2 du &= \int_X^{2X} (t_u^\alpha)' |\zeta'(\sigma + it + it_u^\alpha)|^2 dt_u^\alpha \\ \max_{X \leq u \leq 2X} (t_u^\alpha)' \int_X^{2X} d \left(\int_1^{t+t_u^\alpha} |\zeta'(\sigma + iv)|^2 dv \right) &\leq \max_{X \leq u \leq 2X} (t_u^\alpha)' \int_1^{|t|+t_u^\alpha} |\zeta'(\sigma + iv)|^2 dv \Big|_X^{2X} \\ &\ll_{\sigma} \max_{X \leq u \leq 2X} (t_u^\alpha)' (|t| + t_{2X}^\alpha) \ll_{\sigma, \alpha} X(1 + |t|). \end{aligned}$$

Since $\alpha \in (0, 1]$. Hence, the lemma follows. $\square$

Lemma 16. *Suppose that $\alpha \in (0, 1]$ is a fixed number and $t \in \mathbb{R}$. Then, for fixed $\sigma \in (1/2, 1)$,*

$$\sum_{k=1}^N |\zeta(\sigma + it + it_k^\alpha)|^2 \ll_{\sigma, \alpha} N(1 + |t|).$$

Proof. An application of the ? lemma, see Lemma 1.4 of [1], gives

$$\begin{aligned} \sum_{k=1}^N |\zeta(\sigma + it + it_k^\alpha)|^2 &\ll \int_1^{2N} |\zeta(\sigma + it + it_u^\alpha)|^2 du \\ &+ \left(\int_1^{2N} |\zeta(\sigma + it + it_u^\alpha)|^2 du \int_1^{2N} ((t_u^\alpha)')^2 |\zeta'(\sigma + it + it_u^\alpha)|^2 du \right). \end{aligned}$$

Therefore, it remains to use the estimates of Lemmas 14 and 15. $\square$

Now, we are ready to approximate $\zeta(s)$ by $\zeta_k(s)$.

Lemma 17. *Suppose that α is an arbitrary fixed positive number. Then, for every compact set $K \subset D$*

$$\lim_{n \rightarrow \infty} \limsup_{T \rightarrow \infty} \int_0^T \sup_{s \in K} |\zeta(s + it_u^\alpha) - \zeta_k(s + it_u^\alpha)| du = 0.$$

Proof. Let ϑ be the same as above, and

$$l_n(s) = \frac{s}{\vartheta} \Gamma\left(\frac{s}{\vartheta}\right) n^s,$$

where, as usual, $\Gamma(s)$ is the Euler gamma function. ? that, for $\sigma > 1/2$,

$$\zeta_n(s) = \frac{1}{w\pi i} \int_{\vartheta-i\infty}^{\vartheta+i\infty} \zeta(z+s) l_n(?) dz.$$

? with $\vartheta_0 > 0$ that

$$(25) \quad \zeta_n(s+?) - \zeta(s) \ll \frac{1}{2\pi i} \int_{-\vartheta_0+i\infty}^{-\vartheta_0-i\infty} \zeta(?+s) l_n(z) \frac{dz}{z} + \frac{l_n(1-s)}{1-s}.$$

There exists $\epsilon > 0$ such that the set K has in the strip $\{s \in \mathbb{C} : 1/2 + 2\epsilon \leq \sigma \leq 1 - \epsilon\}$. Now, let

$$\vartheta_0 = \sigma - 1/2 - \epsilon, \quad \vartheta = 1/2 + \epsilon.$$

Then by (25), for all $s \in K$,

$$\begin{aligned} \zeta_n(s + it_u^\alpha) - \zeta(s + it_u^\alpha) &\ll \left| \int_{-\infty}^{\infty} \zeta(1/2 + \epsilon - \sigma + i\tau + \sigma + iv + it_u^\alpha) \frac{l_n(1/2 + \epsilon - \sigma + i\tau)}{1/2 + \epsilon - \sigma + i\tau} \right| \\ &+ \left| \frac{l_n(1-s-it_u^\alpha)}{1-s-it_u^\alpha} \right| \ll \int_{-\infty}^{\infty} |\zeta(1/2 + \epsilon + i\tau + it_u^\alpha)| \\ &\times \sup_{s \in K} \left| \frac{l_n(1/2 + \epsilon - s + i\tau)}{1/2 + \epsilon - s + i\tau} \right| d\tau + \left| \frac{l_n(1-s-it_u^\alpha)}{1-s-it_u^\alpha} \right|. \end{aligned}$$

Hence,

$$(26) \quad \frac{1}{T} \int_0^T \sup_{s \in K} |\zeta_n(s + it_u^\alpha) - \zeta(s + it_u^\alpha)| du \ll I + Z,$$

where

$$I = \int_{-\infty}^{\infty} \left(\frac{1}{T} \int_0^T |\zeta(1/2 + \epsilon + it_u^\alpha + i\tau)| \right) du \sup_{s \in K} \left| \frac{l_n(1/2 + \epsilon - s + i\tau)}{1/2 + \epsilon - s + i\tau} \right| d\tau,$$

and

$$Z = \frac{1}{T} \int_0^T \sup_{s \in K} \left| \frac{l_n(1-s-it_u^\alpha)}{1-s-it_u^\alpha} \right| du.$$

It is well known that, uniformly in $\sigma_1 \leq \sigma \leq \sigma_2$ with ordering $\sigma_1 < \sigma_2$, the estimate

$$\Gamma(\sigma + it) \ll \max\{-c|t|\}$$

with certain $c > 0$ is valid. Therefore, for $s \in K$,

$$\begin{aligned} \frac{l_n(1/2 + \epsilon - s + i\tau)}{1/2 + \epsilon - s + i\tau} &\ll \frac{n^{1/2+\epsilon-\sigma}}{\vartheta} \left| \Gamma\left(\frac{1/2 + \epsilon - \sigma}{\vartheta} + \frac{i(\tau - t)}{\vartheta}\right) \right| \\ (27) \quad &\ll_{\vartheta} n^{-\epsilon} \exp\left\{-\frac{c}{\vartheta}|t - \tau|\right\} \ll_{\epsilon, K} n^{-c} \exp\{-c_1|\tau|\}, \quad c_1 > 0. \end{aligned}$$

This together with Lemma 14 shows that

$$(28) \quad I \ll_{\epsilon, K, \alpha} n^{-\epsilon} \int_{\infty}^{\infty} (1 + |\tau|) \exp\{-c|\tau|\} d\tau \ll_{\epsilon, K, \alpha} n^{-\epsilon}.$$

Reasoning similarly, we obtain that, for some $c_2 > 0$,

$$\begin{aligned} I &\ll_{\epsilon, K} \frac{n^{1/2-2\epsilon}}{T} \int_0^T \exp\{-c_2 t_u^\alpha\} du \\ &\ll_{\epsilon, K} \left(\frac{n^{1/2-\epsilon} \log T}{T} + \frac{n^{1/2-2\epsilon}}{T} \frac{\log T}{T} \exp\{-c_2 t_u^\alpha\} du \right) \\ &\ll_{\epsilon, K} \left(\frac{n^{1/2-\epsilon} \log T}{T} + n^{1/2-2\epsilon} \exp\{-c_2 t_{\log T}^\alpha\} \right) \log T. \end{aligned}$$

Since $t_{\log T}^\alpha \rightarrow \infty$ as $T \rightarrow \infty$, this, (28), and (26) prove the lemma $\square$

The next lemma is a discrete version of Lemma 17.

Lemma 18. *Suppose that $\alpha \in (0, 1]$ is a fixed number. Then, for any compact set $K \subset D$,*

$$\lim_{n \rightarrow \infty} \liminf_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N \sup_{s \in K} |\zeta(s + it_k^\alpha) - \zeta_n(s + it_k^\alpha)| = 0.$$

Proof. Repeating the same arguments as in the proof of Lemma 17, we obtain

$$(29) \quad \frac{1}{N} \sum_{k=1}^N \sup_{s \in K} |\zeta(s + it_k^\alpha) - \zeta_n(s + it_k^\alpha)| \ll \hat{I} + \hat{Z},$$

where

$$\hat{I} = \int_{-\infty}^{\infty} \frac{1}{N} \sum_{k=1}^N |\zeta(1/2 + \epsilon + i\tau + it_k^\alpha)| \sup_{s \in K} \left| \frac{l_n(1/2 + \epsilon - s + i\tau)}{1/2 + \epsilon - s + i\tau} \right| d\tau$$

and

$$\hat{Z} = \frac{1}{N} \sum_{k=1}^N \sup_{s \in K} \left| \frac{l_n(1 - s - it_k^\alpha)}{1 - s - it_k^\alpha} \right|.$$

Therefore, using Lemma 16 and (27), we find the bound

$$(30) \quad \hat{I} \ll_{\epsilon, K, \alpha} n^{-\epsilon}.$$

For $s \in K$, we have

$$\frac{l_n(1 - s - it_k^\alpha)}{1 - s - it_k^\alpha} \ll_{\epsilon, K} n^{1/2-2\epsilon} \exp\{-c_2 t_k^\alpha\}.$$

Therefore,

$$\begin{aligned} \hat{Z} &\ll_{\epsilon, K} \frac{1}{N} \sum_{k=1}^N \exp\{-c_2 t_k^\alpha\} \ll_{\epsilon, K} \frac{n^{1/2-2\epsilon}}{N} \left(\sum_{k \leq \log N} + \sum_{k \geq \log N}^N \right) \\ &\times \exp\{-c_2 t_k^\alpha\} \ll_{\epsilon, K, \alpha} \left(\frac{n^{1/2-2\epsilon}}{N} + n^{1/2-2\epsilon} \exp\{-c_2 t_{\log N}^\alpha\} \right). \end{aligned}$$

This, (29), and (30) prove the lemma. $\square$

Now, we are ready to prove limit theorems 7 and 8.

Proof of Theorem 7. It is more convenient to deal with convergence in distribution in place of weak convergence. Let $\xi_T(?)$ be a random variable defined in a certain probability space with measure μ , and uniformly distributed on $[0, T]$.

Define then $H(D)$ -valued random elements

$$\begin{aligned} X_{T,n} &= X_{T,n}(s) = \zeta_n(s + i\xi_T), \\ Y_T &= Y_T(s) = \zeta(s + i\xi_T), \end{aligned}$$

and denote by $X_n = X_n(s)$ the $H(D)$ -valued random element having the distribution V_n , where $?$. Then, in view of Lemma 11 and the above definitions

$$(31) \quad X_{T,n} \xrightarrow{T \rightarrow \infty} X_n.$$

Moreover, by Lemma 13,

$$(32) \quad X_n \xrightarrow{n \rightarrow \infty} P_\xi.$$

An application of Lemma 17 shows that for every $\epsilon > 0$,

$$\begin{aligned} & \lim_{n \rightarrow \infty} \limsup_{T \rightarrow \infty} \mu\{\rho(Y_T, X_{T,n}) \geq \epsilon\} \\ & \leq \lim_{n \rightarrow \infty} \limsup_{T \rightarrow \infty} \frac{1}{\epsilon T} \int_0^T \rho(\zeta(s + it_u^\alpha), \zeta_n(s + it_u^\alpha)) du = 0, \end{aligned}$$

where ρ is the metric in $H(D)$ inducing its topology of uniform convergence in compacta. The latter equality, $?$, together with Theorem 4.2 of [12] imply the $?$

$$Y_T \xrightarrow{T \rightarrow \infty} P_\xi$$

and the theorem is proved. $\square$

Proof of Theorem 8. We follow the proof of Theorem 7. Let $?$ be a random variable on a certain probability space with measure μ , and having the distribution

$$\hat{\mu}\{\hat{\xi}_N = k\} = \frac{1}{N-1}, \quad k = 2, \dots, N(?)$$

Define the $H(D)$ -valued random elements

$$\hat{X}_{N,n} = \hat{X}_{N,n}(s) = \zeta_n(s + i\hat{\xi}_N) \quad ?$$

and

$$\hat{Y}_N = \hat{Y}_N(s) = \zeta(s + i\hat{\xi}_N),$$

and let X_n be the same random $?$ as in the proof of Theorem 7. Then, by Lemma 12,

$$(33) \quad \hat{X}_{N,k} \xrightarrow{N \rightarrow \infty} X_k.$$

Moreover, Lemma 18 implies that, for every $\epsilon > 0$

$$\begin{aligned} & \lim_{n \rightarrow \infty} \limsup_{N \rightarrow \infty} \hat{\mu}\{\rho(\hat{Y}_N, \hat{X}_{N,n}) \geq \epsilon\} \\ & \leq \lim_{n \rightarrow \infty} \limsup_{N \rightarrow \infty} \frac{1}{\epsilon N} \sum_{k=1}^N \rho(\zeta(s + it_k^\alpha), \zeta_n(s + it_k^\alpha)) = 0. \end{aligned}$$

Therefore, (33), (32) and Theorem $?$, and the theorem is proved. $\square$

4. PROOF OF UNIVERSALITY

Proofs of Theorems ?? and 6 are standard, and ? of Theorems 7 and 8 as well as the Mergelyan theorem ? the approximation of analytic functions by polynomials [16].

Proof of Theorem ??. By the Mergelyan theorem, there exists a polynomial $p(s)$ such that

$$(34) \quad \sup_{s \in K} |f(s) - e^{p(s)}| < \frac{\epsilon}{2}.$$

In view of Lemma 13, the function $e^{p(s)}$ belongs to the support of the measure P_ξ . Therefore

$$(35) \quad P_\xi(G_\epsilon) > 0,$$

where

$$G_\epsilon \{g \in H(D) : \sup_{s \in K} |g(s) - e^{p(s)}| < \epsilon/2\}.$$

Thus, Theorem 7 and the equivalent of weak convergence in ? set imply the inequality

$$\liminf_{T \rightarrow \infty} P_T(G_\epsilon) \geq P_\xi(G_\epsilon) > 0.$$

Now, the definitions of P_ξ and G_ϵ together with (34) prove the first assertion of the theorem.

To prove the second assertion of the theorem, define the set

$$\hat{G}_\epsilon = \{g \in H(D) : \sup_{s \in K} |g(s) - f(s)| < \epsilon\}.$$

The latter set is a continuity set of the measure P_ξ for all but at most countably many $\epsilon > 0$. Therefore, Theorem 7 and the equivalent of weak convergence in terms of continuity sets show that

$$(36) \quad \lim_{T \rightarrow \infty} P_T(\hat{G}_\epsilon) = P_\xi(G_\epsilon)$$

for all but at most countably many $\epsilon > 0$. In view of (34), we have the inclusion $G_\epsilon \subset \hat{G}_\epsilon$. Therefore, $P_\xi(\hat{G}_\epsilon) > 0$ by (35) and the definitions of P_T and $\hat{G}_\epsilon$, and (36) prove the second assertion of the theorem. $\square$

Proof of Theorem 6. uses Theorem 8 in place of Theorem 7, and follows that of Theorem ??. $\square$

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